

*Measuring multipartite entanglement through dynamic susceptibilities*Philipp Hauke,^{1,2,*} Markus Heyl,^{1,2,3} Luca Tagliacozzo,^{4,5} and Peter Zoller^{1,2}¹*Institute for Theoretical Physics, University of Innsbruck, 6020 Innsbruck, Austria*²*Institute for Quantum Optics and Quantum Information of the Austrian Academy of Sciences, 6020 Innsbruck, Austria*³*Physik Department, Technische Universität München, 85747 Garching, Germany*⁴*ICFO-Institut de Ciències Fotoniques, Mediterranean Technology Park, 08860 Castelldefels (Barcelona), Spain*⁵*Department of Physics and Scottish Universities Physics Alliance University of Strathclyde, Glasgow G4 0NG, Scotland, UK*

In this supplementary material, (1) we discuss the relationship Fisher-information/susceptibility in a classical, thermal setting; (2) we present a sum rule to bound the quantum Fisher information; (3) we explain in detail the scaling behaviour for the quantum Fisher information; (4) we present additional numerical results for universal scaling: (4.A) a quantum phase transition above the upper critical dimension in the infinite-range Ising model, (4.B) a transition coinciding with a factorization point in one-dimensional hard-core bosons in an optical lattice, and (4.C) a symmetry-protected topological transition in the Kitaev wire.

S1. CLASSICAL CONTRIBUTION TO QFI AND THERMAL PHASE TRANSITIONS

In the main text, we study how well thermal states that lie at close points in a phase diagram are distinguishable, focusing entirely on quantum effects. More precisely, we ask how fast the probability distribution of a thermal state $\rho = \sum_{\lambda} p_{\lambda} |\lambda\rangle \langle \lambda|$, with $p_{\lambda} = \exp(-E_{\lambda}/T)/Z$, changes due to the unitary transformation with $e^{i\vartheta\hat{O}}$. It is instructive to extend this analysis to any source of distinguishability, be it quantum or classical.

One may extend the present scenario by coupling the system to an external bath and including a sufficiently long waiting time, such that the populations can rethermalise. The changing parameter ϑ may then signify the temperature. The change of populations that this induces provides a second source of distinguishability between ρ and ρ' , which to leading order remains disparate from the one described above. We can incorporate this effect by considering the full quantum Fisher information (QFI) [1, 2]

$$\begin{aligned} \tilde{F}_Q &= F_C + F_Q \\ &= \sum_{\lambda} \frac{(\partial p_{\lambda}/\partial \vartheta)^2}{p_{\lambda}} + 2 \sum_{\lambda, \lambda'} \frac{(p_{\lambda} - p_{\lambda'})^2}{p_{\lambda} + p_{\lambda'}} \left| \left\langle \lambda \left| \frac{\partial \lambda'}{\partial \vartheta} \right\rangle \right|^2. \end{aligned} \quad (\text{S1})$$

The first term, F_C , cannot appear under purely unitary transformations. The second term, F_Q , the subject of this work, is the term that is relevant at a QPT at $T = 0$, where the first term, F_C , vanishes identically.

Both terms are related to different aspects of the susceptibility of ρ towards the generator $\hat{O}$. As demonstrated in Eq. (4) of the main text, F_Q is in one-to-one correspondence to the Kubo dynamic susceptibility at non-zero frequencies. In Kubo's linear response theory, a system is perturbed weakly in a time-dependent way,

generating a unitary dynamics, but—to leading order—without changing the populations of energy levels [3].

The change of probabilities, instead, is related to the static, isothermal susceptibility, as has already been pointed out within purely classical statistical physics [4]. Here, we demonstrate this relation in a quantum statistical calculation. Consider a static perturbation, $H \rightarrow H' = H + \vartheta' \hat{O}$, with $\vartheta' = T\vartheta$ (i.e., we choose the energy scale $\mathcal{J} = T$). The expectation value of the observable $\hat{O}$ is given by a derivative of the free entropy $\ln Z$, with $Z = \sum_{\lambda} \exp(-E_{\lambda}/T)$ the partition function,

$$\langle \hat{O} \rangle = T \frac{\partial \ln Z}{\partial \vartheta'} = - \sum_{\lambda} p_{\lambda} \frac{\partial E_{\lambda}}{\partial \vartheta'}. \quad (\text{S2})$$

(Here and in the following, we understand derivatives with respect to ϑ' to be evaluated at $\vartheta' = 0$.) The corresponding static, isothermal susceptibility is

$$\chi^T \equiv \frac{\partial \langle \hat{O} \rangle}{\partial \vartheta'} = - \sum_{\lambda} \frac{\partial p_{\lambda}}{\partial \vartheta'} \frac{\partial E_{\lambda}}{\partial \vartheta'} - \sum_{\lambda} p_{\lambda} \frac{\partial^2 E_{\lambda}}{(\partial \vartheta')^2}. \quad (\text{S3})$$

The derivative of p_{λ} reads

$$\frac{\partial p_{\lambda}}{\partial \vartheta'} = - \frac{p_{\lambda}}{T} \left(\frac{\partial E_{\lambda}}{\partial \vartheta'} - \sum_{\lambda'} p_{\lambda'} \frac{\partial E_{\lambda'}}{\partial \vartheta'} \right) \quad (\text{S4})$$

One can use perturbation theory to evaluate the change of energy levels following the static perturbation. Working in a basis where degenerate eigenlevels $|\lambda\rangle$ are prediagonalised with respect to the perturbation operator $\hat{O}$, we obtain

$$\begin{aligned} \chi^T &= \frac{1}{T} \left(\sum_{\lambda} p_{\lambda} \hat{O}_{\lambda, \lambda}^2 - \langle \hat{O} \rangle^2 \right) + \sum_{\lambda, \lambda'}^{E_{\lambda} \neq E_{\lambda'}} \frac{|\hat{O}_{\lambda, \lambda'}|^2 (p_{\lambda} - p_{\lambda'})}{E_{\lambda'} - E_{\lambda}} \\ &= \chi_{\text{el}} + \chi_{\text{vV}}. \end{aligned} \quad (\text{S5})$$

Here, we defined the abbreviation $\hat{O}_{\lambda, \lambda'} \equiv \langle \lambda | \hat{O} | \lambda' \rangle$. The first term is the elastic contribution, or classical Curie

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contribution, arising from a change of populations. It vanishes in a pure state, such as the quantum ground state at $T = 0$. The second term is the quantum-mechanical van-Vleck correction, arising from matrix elements between different energy levels. It is connected smoothly to the real part of the Kubo dynamic susceptibility at non-zero frequencies and vanishes in a purely classical theory. Importantly, the elastic peak is a singular contribution at exactly $\omega = 0$, and does thus not influence F_Q . Since it is this elastic peak that diverges at thermal phase transitions, these do not appear as particular features in F_Q .

We can connect the elastic peak to the classical contribution to the Fisher information [4]. Equation (S4) yields $F_C = \sum_{\lambda} \frac{(\partial p_{\lambda}/\partial \theta)^2}{p_{\lambda}} = \chi_{\text{el}}/T$. Similarly to χ^T , $\tilde{F}_Q$ thus has two parts: $\tilde{F}_Q$, the subject of this article, related to entanglement and divergent at QPTs; and F_C , of non-unitary, classical origin and divergent at thermal phase transitions. Both are related to the order parameter's susceptibility. Remarkably, by choosing the susceptibility at vanishing or non-vanishing frequencies, we can extract the purely classical or the purely quantum-mechanical contribution to the QFI.

S2. SUM RULE

Our central relation Eq. (2) allows us to derive a Thomas-Reiche-Kuhn-type [5–7] sum rule that may be useful to bound the QFI. Since $\tanh(\frac{\omega}{2T}) \leq \frac{\omega}{2T}$, we have

$$F_Q(T) \leq \frac{1}{T} \frac{2}{\pi} \int_0^{\infty} d\omega \omega \chi''(\omega, T). \quad (\text{S6})$$

Due to the Kramers–Kronig relations, we can now take advantage of a superconvergence theorem [8]. This theorem states that two complex functions g and f , for which $g(x) = P \int_0^{\infty} d\tilde{x} \frac{f(\tilde{x})}{x^2 - \tilde{x}^2}$ holds, are related via $\int_0^{\infty} dx f(x) = \lim_{x \rightarrow \infty} (x^2 g(x))$, with P the principal part of the integral. By setting $f(x) = -\frac{2}{\pi} x \chi''(x)$ and $g(x) = \chi'(x) \equiv \Re(\chi(x))$, one obtains

$$F_Q(T) \leq -\frac{1}{T} \lim_{\omega \rightarrow \infty} (\omega^2 \chi'(\omega, T)), \quad (\text{S7})$$

a Thomas-Reiche-Kuhn-type sum rule that upper bounds the QFI. Here, as in Eq. (4) of the main text, the limit to infinity is to be understood as a frequency range much larger than all energy scales of the considered degrees of freedom (but smaller than energy scales at which resonances to additional degrees of freedom appear, the entanglement between which, however, could be analysed analogously). It is interesting to note that, if we take the classical limit by sending $T \rightarrow \infty$, the bound (S7) becomes an equality relating the QFI to the high-frequency limit of $\chi'(\omega)$.

S3. SCALING AND UNIVERSALITY

In the main text, we have demonstrated a direct relationship of the QFI to thermodynamic response functions. These response functions underlie universal scaling behaviour close to quantum critical points, and consequently directly imprint a universal scaling behaviour onto the QFI. Aspects of critical scaling of the QFI and related quantum metrics have also been discussed, e.g., in Refs. [9–16]. The purpose of the present supplementary section is to pedagogically present the scaling theory for the QFI.

A. Basic principles

Before discussing the scaling of the QFI in detail, let us briefly recall the basic principles of the scaling Ansatz [17]. Here, we focus on lattice models with lattice spacing a that have a continuous phase transition at zero temperature, *viz* a quantum phase transition (QPT). Generalisation to continuum models is straightforward. The dimensionality of the system is d , its linear size L , and the total particle number $N = L^d$ (where we set $a = 1$). To simplify the discussion, we focus on the simplest scenario, QPTs obtained by tuning a single relevant field, say the magnetic field h , to the critical point h_c . Sufficiently close to the phase transition, very diverse microscopic models can develop the same collective emerging behaviour at large length scales, i.e., they display the same power-law decay of correlations, which is described by a small set of universal exponents $\{\Delta_{\alpha}\}$. Such a universality emerges between microscopic Hamiltonians if they differ only by terms (called irrelevant operators) that vanish under coarse-graining transformations on the lattice. The corresponding Hamiltonians become more and more similar in the infra-red limit, and the emerging physics becomes the same.

In finite systems, or systems at finite temperature, the power-law of the correlations is cut off by system size or the thermal correlation length. In such cases, one can extract the universal properties of the QPT by performing an appropriate analysis of scale transformations, i.e., transformations that modify the lattice spacing $a \rightarrow \lambda a$, where $\lambda > 1$ (this scale transformation can be envisioned as grouping subsets of λ lattice sites together). At the critical point one can expand all local operators in the basis of scaling operators, i.e., those that transform under rescaling with a simple power of λ , $\hat{\mathcal{O}}_i^{\alpha} \rightarrow \lambda^{-\Delta_{\alpha}} \hat{\mathcal{O}}_i^{\alpha}$, with scaling exponent $\Delta_{\alpha} \geq 0$. The two-point correlation function of these operators decays as

$$\langle \hat{\mathcal{O}}_0^{\alpha} \hat{\mathcal{O}}_r^{\alpha} \rangle \propto r^{-2\Delta_{\alpha}}. \quad (\text{S8})$$

In particular, the fixed-point Hamiltonian is a scaling operator with scaling exponent Δ_H .

Scaling operators that transform with $\Delta_{\alpha} > \Delta_H$ are called *irrelevant* operators, since, when added as a weak

perturbation, coarse graining decreases their importance relative to the original Hamiltonian. Those operators with $\Delta_\alpha < \Delta_H$, on the other hand, are called *relevant*, since they become larger and larger under coarse-graining transformations, eventually driving the system to a different fixed point. In the *scaling regime*, the regime where relevant perturbations are still small compared to the original Hamiltonian, one can extract the universal low-energy physics by studying the response of the system to scaling transformations.

Relevant perturbations are, e.g., the inverse system size L^{-1} , the finite temperature T [18], and the normalized distance from the critical point $\tilde{h} = |h - h_c|/h_c$. They increase under a rescaling as $L^{-1} \rightarrow L'^{-1} = \lambda L^{-1}$, $T \rightarrow T' = \lambda^z T$, and $\tilde{h} \rightarrow \tilde{h}' = \lambda^{1/\nu} \tilde{h}$, where z is the dynamical critical exponent and ν the correlation-length critical exponent. The scaling regime for a finite and cold system is thus defined by $L^{-1}, T \ll 1$, where we can consider inverse system size and temperature as small perturbations and project the full theory on the scaling operators, the eigenvectors of the linearized scaling transformation close to the critical point. We will now discuss the scaling behaviour of the QFI with respect to these perturbations.

B. Scaling of the quantum Fisher information

As the starting point, we first discuss the scaling of the ground-state QFI in the vicinity of a quantum phase transition [12–15]. Afterwards, we will give the generalization to non-zero temperatures. In the ground state, the QFI related to a scaling operator, $\hat{O} = \sum_l \hat{O}_l^\alpha$, takes the simple form [19]

$$f_Q = \frac{4}{L^d} \sum_l \sum_r \langle \hat{O}_l^\alpha \hat{O}_{l+r}^\alpha \rangle. \quad (\text{S9})$$

We want to understand how it transforms under scaling transformations. On the lattice, a scaling transformation of a factor $\lambda > 1$ implies grouping λ sites together in a single block. Under a scale transformation $L \rightarrow L' = L/\lambda$, $l \rightarrow l' = l/\lambda$, $r \rightarrow r' = r/\lambda$. Remembering that scaling operators transform as $\hat{O}_l^\alpha \rightarrow \lambda^{-\Delta_\alpha} \hat{O}_l^\alpha$, if we assume that the ground state is invariant under such a transformation and that the correlation function varies slowly inside each block, we obtain

$$f_Q = \lambda^{d-2\Delta_\alpha} \frac{4}{L'^d} \sum_{l'} \sum_{r'} \langle \hat{O}_{l'}^\alpha \hat{O}_{l'+r'}^\alpha \rangle. \quad (\text{S10})$$

If we now define the scaling exponent for the QFI

$$\Delta_Q = d - 2\Delta_\alpha, \quad (\text{S11})$$

the QFI transforms as

$$f_Q \rightarrow \lambda^{\Delta_Q} f_Q. \quad (\text{S12})$$

(note the similarity to the scaling of the geometric tensor of eigenstates, which has scaling exponent $\Delta_Q + 2z$ [9, 10]).

One can derive this scaling behaviour also at non-zero temperatures starting from our central formula (4), which is valid at any T . To this end, we consider the dynamic structure factor

$$S(\omega) = \frac{1}{2} \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \hat{O} \hat{O}(t) + \hat{O}(t) \hat{O} \rangle, \quad (\text{S13})$$

where $\hat{O}(t) = e^{iHt} \hat{O} e^{-iHt}$. Inserting $S(\omega)$ into Eq. (4) by use of the fluctuation–dissipation relation $\chi''(\omega) = \tanh(\omega/2T) S(\omega)$, the QFI density becomes

$$f_Q = \frac{2}{\pi L^d} \int_{-\infty}^{\infty} d\omega \tanh^2\left(\frac{\omega}{2T}\right) S(\omega). \quad (\text{S14})$$

Since under a scale transformation frequencies and temperatures scale in the same way [20], $T \rightarrow \lambda^z T$ and $\omega \rightarrow \lambda^z \omega$, the above formula has the same scaling behaviour as $2 \int_{-\infty}^{\infty} d\omega S(\omega) / (\pi L^d) = 4 \langle \hat{O}^2 \rangle / L^d$. We are thus back at Eq. (S9), and re-obtain the scaling behaviour (S12).

In a large but finite system, close to the critical point, and for temperatures close to zero, considering how $L, T, \tilde{h}$ transform under scale transformations, we thus expect

$$f_Q(T, L^{-1}, \tilde{h}) = \lambda^{\Delta_Q} \phi_Q(\lambda^z T, \lambda L^{-1}, \lambda^{1/\nu} \tilde{h}) + C. \quad (\text{S15})$$

The constant C is related to the non-universal short-distance correlations. The scaling function ϕ_Q encodes the large-distance part of the correlation functions, which dominates the scaling if the condition $\Delta_Q > 0$ is fulfilled. In the following discussion, we assume that this is the case, allowing us to neglect the non-universal constant C , leading to Eq. (7). Below we will encounter, in the guise of the hard-core boson chain, an example where $\Delta_Q \leq 0$. In such cases, derivatives of f_Q still allow one to extract the purely universal contribution. Note that when $\Delta_Q = 0$ there can be logarithmic corrections to Eq. (S15).

We can use the scaling form Eq. (S15) as long as all relevant perturbations are small, that is as long as we stay in the scaling regime. The most relevant perturbation will constitute the most significant breaking of scale invariance, a fact that enables us to extract valuable information about the involved critical exponents. For example, when both L^{-1} and T are sufficiently small, i.e., if $L^{-1}, T^{1/z} < \tilde{h}^\nu$, the deviation $\tilde{h}$ from the critical point represents the most relevant perturbation: scale invariance will be broken at the scale $\lambda \sim \tilde{h}^{-\nu}$. This implies that we can trade λ by $\tilde{h}^{-\nu}$, and varying $\tilde{h}$ in this regime has the same effect as a rescaling transformation. In this regime, we expect f_Q therefore to behave as

$$f_Q(T, L^{-1}, \tilde{h}) = \tilde{h}^{-\nu \Delta_Q} \phi_{Q1}\left(T \tilde{h}^{-\nu z}, L^{-1} \tilde{h}^{-\nu}\right). \quad (\text{S16})$$

For sufficiently relevant operators, for which $\Delta_Q > 0$, f_Q thus diverges as $\tilde{h}^{-\nu\Delta_Q}$ when approaching the critical point. In the above expression, $\phi_{Q1}(T\tilde{h}^{-\nu z}, L^{-1}\tilde{h}^{-\nu})$ is a scaling function, embodying the fact that the normalised QFI density, $f_Q/\tilde{h}^{-\nu\Delta_Q}$, is a well defined function of the scaling variables $T\tilde{h}^{-\nu z}$ and $L^{-1}\tilde{h}^{-\nu}$ rather than of L , T , and $\tilde{h}$ separately.

Upon decreasing $\tilde{h}$, the system will reach a crossover with the next relevant perturbation. If $L^{-1} < T^{1/z}$, the crossover will occur around $T^{1/z} \sim \tilde{h}^\nu$ and carry over into a thermally dominated regime, for which the scaling of the QFI is dictated by

$$f_Q(T, L^{-1}, \tilde{h}) = T^{-\Delta_Q/z} \phi_{Q2}(L^{-1}T^{-1/z}, \tilde{h}T^{-1/(z\nu)}) . \quad (\text{S17})$$

This scaling behaviour characterises the robustness of critical entanglement against non-zero temperatures (see, e.g., Ref. [21] for a discussion of the robustness in presence of noise).

In the opposite case $L^{-1} > T^{1/z}$, upon reducing $\tilde{h}$ one enters a regime dominated by finite-size effects. Here, f_Q behaves as

$$f_Q(T, L^{-1}, \tilde{h}) = L^{\Delta_Q} \phi_{Q3}(TL^z, L^{1/\nu}\tilde{h}) . \quad (\text{S18})$$

In this regime, f_Q saturates to a finite, L -dependent value, which diverges with system size as L^{Δ_Q} . The crossover is located around $L^{-1} \sim \tilde{h}^\nu$ and should thus approach the critical value $\tilde{h} = 0$ as $L^{-1/\nu}$. Another interesting scenario appears when T and $1/L$ are comparable, giving rise to further complex crossover phenomena. This can be seen, e.g., in Fig. 2 of the main text, where we study the universal scaling behaviour of the QFI in the paradigmatic model of the quantum Ising chain.

C. Multipartite entanglement length

This scaling analysis immediately permits us to identify strongly entangled quantum critical points. These are those critical points where $\Delta_Q > 0$, i.e., where at low temperatures and close to criticality, f_Q diverges with system size. Additionally, the scaling also allows us to extract a length scale of the entanglement present in the system [22]. Combining Eq. (S15) and main-text inequality (1), one sees that the number of entangled particles scales as $N_Q \geq f_Q \sim \lambda^{d-2\Delta_Q}$, which defines a multipartite entanglement length scale

$$l_{\text{ent}} \sim N_Q^{1/d} \gtrsim \lambda^{1-2\Delta_Q/d} . \quad (\text{S19})$$

Here, we assumed an isotropic distribution of entanglement; for anisotropic situations, the longest distance along which particles are witnessed to be entangled is actually larger. This entanglement length is related to the correlation length ξ of the generator $\hat{\mathcal{O}}$. Since at criticality the correlation length scales as $\xi \sim \lambda$, one has $l_{\text{ent}} \sim \xi^{1-2\Delta_Q/d}$. For $2\Delta_Q/d < 1$, the length scale over

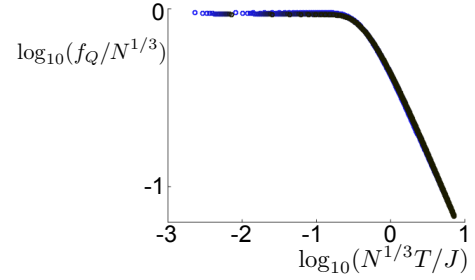


Figure S1. Universal scaling of the QFI density, $f_Q = F_Q/N$, calculated for the order parameter in the quantum Ising model with infinite-range interactions. Finite-temperature and finite-size scaling at criticality $\gamma = \pi/4$. The bullets denote exact data, from lighter to darker blue for $N = 100, \dots, 3000$. The universal scaling laws produce an excellent data collapse. The divergence $f_Q \sim N^{1/3}$ is much weaker than in the one-dimensional quantum Ising model, reflecting the more mean-field-like character of the infinite-range model. Nevertheless, it still implies a diverging multipartiteness of entanglement.

which particles are entangled with each other thus diverges with system size.

S4. EXAMPLES FOR UNIVERSAL SCALING

In the main text, we illustrate the universal scaling of the QFI using one of the most simple and paradigmatic models hosting a quantum phase transition, the Ising model in a transverse field. In this section, we spotlight additional aspects of the universal behaviour of the QFI, exemplified in three different models. First, we briefly provide the scaling analysis for the quantum Ising model with infinite-range interactions capturing the universal properties of the Ising transition above the upper critical dimension (Sec. S4 A). Then, we study the Mott-insulator–superfluid transition of hard-core bosons to show that relevant operators other than the order parameter also exhibit universal behaviour (Sec. S4 B). Finally, we analyze the QFI in the Kitaev wire, a model that has a topological quantum phase transition (Sec. S4 C). We show that while the QFI does not witness strong entanglement in this model, it is nevertheless sensitive to the topological transition.

A. The infinite-range quantum Ising model

In the main text, we considered the infinite-range Ising model in a transverse field, main-text Eq. (8) with $\mathcal{J}_{ij} = 1/N$, to illustrate the absence of features in the QFI at thermal transitions (see Fig. 3). Here, we provide its scaling analysis in the vicinity of its quantum phase transition at non-zero temperatures, using exact numerical results (see Methods). Scaling behaviour of this model in its ground-state has been discussed, e.g., in

Refs. [12, 23].

The infinite-range Ising model describes the universal properties above the upper critical dimension, with the longitudinal magnetization $\hat{O} = \sum_{l=1}^N \sigma_l^x / 2$ as the order parameter. In Fig. S1, we plot the QFI as a function of temperature for different system sizes at the system's quantum critical point, $\tilde{h} = 0$. Using that $\Delta_Q = 1/3$ and $z = 1/3$ [24] one obtains the scaling of the QFI density $f_Q \sim N^{1/3} \phi(TN^{1/3})$. From the scaling collapse, we determine numerically that $\Delta_Q = 0.34$ and $z = 0.33$, which is in very good agreement with the exact solution.

B. Mott-insulator–superfluid transition of hard-core bosons

The scaling behaviour of f_Q , Eq. (S15), appears in all relevant operators, not only the order parameter. As an illustration for this fact, we choose a system that is relevant for experiments on cold gases, namely a system of ultracold bosonic atoms confined in a one-dimensional optical lattice. In the limit of strong on-site interactions, this system can be modelled by the Hamiltonian [25]

$$H_{\text{hcb}} = -J \sum_{l=1}^{N-1} (\hat{b}_l^\dagger \hat{b}_{l+1} + \text{h.c.}) - 2\mu \sum_{l=1}^N \hat{b}_l^\dagger \hat{b}_l, \quad (\text{S20})$$

where the bosonic creation (annihilation) operators $\hat{b}_l^\dagger$ ($\hat{b}_l$) obey the hard-core constraint $\hat{b}_l^\dagger \hat{b}_l^\dagger = 0$. This model has a QPT at $\mu_c/J = 1$ from a critical superfluid phase to a Mott-insulator at uniform filling, with critical exponents $z = 2$ and $\nu = 1/2$ [20].

The transition has several peculiarities. First, the exact ground state is fully separable at the QPT [26]. Second, one of its order parameters, the particle density $\hat{n} = \sum_l \hat{b}_l^\dagger \hat{b}_l / N$, commutes with the Hamiltonian. To get a non-zero signal in the QFI, we can, however, use a modified observable that is still a relevant operator, the staggered density $\hat{O} = \sum_l (-1)^l \hat{b}_l^\dagger \hat{b}_l$. In optical-lattice experiments, the dynamic susceptibility of such density fluctuations may be probed by Bragg spectroscopy [27, 28].

The Hamiltonian Eq. (S20) is exactly solvable with the same methods as for the Ising chain in a transverse field [20, 29]. Moreover, under periodic boundary conditions, the dynamic structure factor [see Eq. (S13)] associated to density fluctuations $\hat{O} = \sum_l e^{iqr_l} \hat{b}_l^\dagger \hat{b}_l$ at wave vector q can be evaluated analytically [30]. Via the fluctuation–dissipation relation, the dynamic structure factor in turn determines the dynamic susceptibility. Specifying the argument q explicitly, one finds that $S(q, \omega)$ is gapless at $q = 0$ and acquires a maximum at $q = \pi$, with

$$S(\pi, \omega) = 2\pi \sum_k \delta(\omega - \epsilon_k + \epsilon_{k+\pi}) f(\epsilon_{k+\pi}) [1 - f(\epsilon_k)], \quad (\text{S21})$$

with $\epsilon_k = -2[J + \mu \cos(k)]$ the single-particle energy of the diagonalised Hamiltonian and $f(\epsilon_k) =$

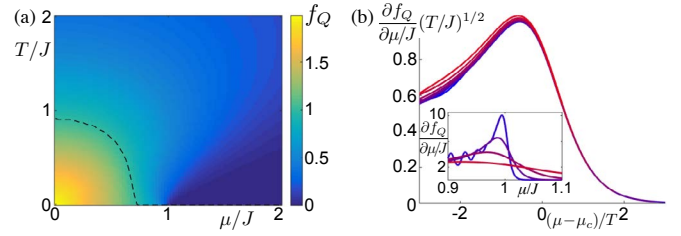


Figure S2. Quantum Fisher information density, f_Q , for the staggered density of hard-core bosons in a one-dimensional optical lattice. (a) In part of the superfluid phase, f_Q witnesses two-site entanglement. Exact data for $N = 1000$. (b) Inset: $\partial f_Q / \partial \mu$ shows a peak that, with decreasing T , diverges and moves towards the critical point. Main panel: The predicted scaling exponents produce a perfect data collapse for $\partial f_Q / \partial \mu$. Exact data for $N = 100$ and, from blue to red, 8 different temperatures distributed on a logarithmic scale between $T = 0.01 - 0.2J$ (inset: 4 temperatures in $0.01 - 0.13J$).

$1/[\exp(\epsilon_k/T) + 1]$ the Fermi–Dirac occupation. As a consequence of the gaplessness, the QFI evaluated at $q = 0$ vanishes, while it assumes its maximum value at $q = \pi$, i.e., for the staggered density. Using the fluctuation–dissipation theorem and evaluating the integral in Eq. (4), the corresponding QFI density becomes

$$f_Q = \frac{4}{N} \sum_k \tanh^2 \left(\frac{\epsilon_k - \epsilon_{k+\pi}}{2T} \right) f(\epsilon_{k+\pi}) [1 - f(\epsilon_k)]. \quad (\text{S22})$$

This quantity is plotted in Fig. S2(a) as a function of temperature and chemical potential, for $N = 1000$. The entanglement witnessed by the QFI extends up to $T \approx 0.91J$, improving over other witnesses showing entanglement up to $T \approx 0.7J$ [31, 32]. In contrast to Ref. [31], however, the QFI for the staggered magnetisation does not witness genuine multipartiteness of entanglement, but saturates at the value $f_Q = 2$. (Notably, the QFI for other generators such as $b_l + b_l^\dagger$ does show divergent behaviour.) Towards the phase transition, the QFI for the staggered density decreases, consistent with the presence of the factorisation point. Despite this absence of divergences, the phase transition imprints its character onto the universal behaviour of the QFI and its derivatives, as explained in what follows.

At $T = 0$ and $N = \infty$, one can use Eq. (2) to derive a particularly simple analytical formula for the QFI of the staggered density,

$$f_Q = \begin{cases} 4n & \mu \leq 0 \\ 4(1-n) & \mu > 0 \end{cases}. \quad (\text{S23})$$

Here, we defined the mean particle density $n \equiv \langle \hat{n} \rangle$, which is $n = 1 - \text{acos}(\mu/J)/\pi$ for $|\mu| < J$ and 0 else. The QFI density attains its maximal value of $f_Q = 2$ at $\mu = 0$ and decreases towards the phase-transition point, where it vanishes. The above analytic expression allows us to obtain the scaling dimension of f_Q by expanding it close to the critical point, $f_Q \propto \tilde{h}^{1/2}$,

with $\tilde{h} = 1 - \mu/J$ and $\mu/J \leq 1$. Comparison with the scaling form Eq. (S16) gives the exact scaling exponent $\Delta_Q = -1$. Since $\Delta_Q < 0$, f_Q of the staggered magnetisation displays no divergent behaviour.

Nevertheless, as seen in the inset of Fig. S2(b), the non-analyticity of f_Q at $\tilde{h} = 0$ leads to a divergence in its derivatives. Considering the temperature as the most relevant perturbation, Eq. (S17), we obtain the universal behaviour

$$\frac{\partial f_Q}{\partial \tilde{h}} = (T/J)^{-\Delta_Q/z+1/(z\nu)} \phi_{Q3} \left(\tilde{h} (T/J)^{-1/(z\nu)} \right). \quad (\text{S24})$$

That is, $\partial f_Q / \partial \tilde{h}$ has a peak that diverges as $(T/J)^{-\Delta_Q/z+1/(z\nu)}$ and the position of which lies at a line $\tilde{h} \sim (T/J)^{1/(z\nu)}$. Using the analytically known exponents $-\Delta_Q/z + 1/(z\nu) = 1/2$ and $1/(z\nu) = 1$, an excellent scaling collapse is achieved up to rather large temperatures [main panel of Fig. S2(b)]. We also performed fits to the numerical data for the peak height and peak position. These yield exponents of 0.49 and 1.02, respectively—even without high-precision data, we are able to reproduce the exact scaling exponents within two percent accuracy.

C. Topological transition in the Kitaev wire

Entanglement, in particular the entanglement entropy, has proven important for the characterization of topological systems [33]. The natural question thus arises whether the QFI as an entanglement witness is capable to quantify and characterize topological phases or topological phase transitions. On a fundamental level, topological order cannot be captured by local observables that appear in conventional dynamic susceptibilities such as considered for the QFI in the main text. Still, the QFI is, remarkably, capable to detect topological phase transitions, as we will see in the example of the Kitaev chain studied in the following. It will be an interesting research question whether the QFI or non-local variations thereof might be useful for quantifying topological phases.

One of the simplest examples of a topological transition is given by the Kitaev wire [34], a one-dimensional model of free fermions with both tunnelling and pairing,

$$H_{\text{Kw}} = -J \sum_{l=1}^{N-1} \left(\hat{c}_l^\dagger \hat{c}_{l+1} + \Delta \hat{c}_l^\dagger \hat{c}_{l+1}^\dagger + \text{h.c.} \right) + 2\mu \sum_{l=1}^N \hat{c}_l^\dagger \hat{c}_l. \quad (\text{S25})$$

Here, a fermion at site l is created (annihilated) by $\hat{c}_l$ ($\hat{c}_l^\dagger$). Being a free-fermion model, this model can be solved exactly [35, 36]. For $\Delta = 1$, its eigenenergies coincide with those of the Ising chain in a transverse field [20], which is studied in the main text. Importantly, however, although the spectra are identical, the Jordan–Wigner transformation that establishes the mapping is highly non-local. Thus, operators or correlation functions that are local

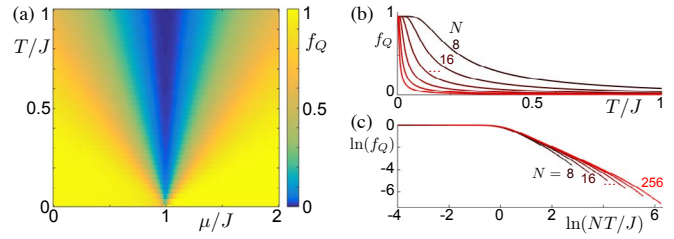


Figure S3. Quantum Fisher information density, f_Q , for non-interacting fermions in a one-dimensional Kitaev wire with equal tunnelling and pairing, for the generator $\hat{O} = \sum_i (\hat{c}_i + \hat{c}_i^\dagger)/2$. (a) f_Q does not witness entanglement, but it is sensitive to the presence of the topological quantum critical point. Exact data for $N = 128$. (b) Cuts along temperature, at the critical point, $\mu/J = 1$. The sensitivity of f_Q increases with system size. (c) A perfect data collapse is achieved with the predicted critical exponents. Deviations appear where T leaves the critical regime, i.e., where the non-universal high-temperature behaviour sets in. Data in panels b and c, from dark to light red: $N = 8, 16, \dots, 256$.

in the Ising chain do not necessarily transform into local objects in the Kitaev model and vice versa. For example, the conventional symmetry-broken phase in the Ising model can on the Kitaev side only be captured by string order parameters, i.e., by non-local correlation functions. This phase is separated from a trivial phase at large chemical potential by a topological transition.

In the following, we set the pairing strength $\Delta = 1$ and tune the chemical potential μ to access the topological transition, choosing the generator $\hat{O} = \sum_i (\hat{c}_i + \hat{c}_i^\dagger)/2$ to study it. This operator is only local for the Kitaev chain; if we were to use the Jordan–Wigner transformation to convert into spin language, it would become a sum of non-local operators. Thus, the associated dynamic susceptibility defining the QFI is only meaningful in the Kitaev model but not the Ising chain. Figure S3 presents the corresponding QFI density in the chemical-potential–temperature plane. Its value does not exceed $f_Q = 1$; contrary to other cases analyzed above, the QFI does not witness entanglement. Importantly, however, as Fig. S3 shows, the QFI is able to detect the presence of the topological phase transition in the Kitaev wire. This behaviour is not only captured by the considered observable, but also by other generators not shown here.

What is more, the QFI displays clear universal behaviour, which we analyze analogous to Fig. 2 of the main text, i.e., by studying cuts along T for different system sizes, with the chemical potential fixed to the critical value. Since the transition is of Ising type, it has the critical exponents $\nu = z = 1$. Additionally, as shown by Fig. S3(b), the QFI density approaches $f_Q = 1$ at $T = 0$, i.e., we have $\Delta_Q = 0$. In consequence, we expect from Eq. (S17) or (S18) f_Q to be a universal function of TN within in the critical regime. This expectation is confirmed by the scaling collapse displayed in Fig. S3(c).

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